General Science

Alveary

General Science: Grade 8

GRADE(S): 8

About the Course

This is the second of a two-year Physical Science course (i.e., introductory physics and chemistry). The fullness of the course includes topics in earth science, history, politics, and more. The completion of this course provides a gentle yet robust transition into the high school disciplines. The lab component is required for this course. Coordinating afternoon activities are provided in Outdoor Work.

This topic is included in the following course(s): Science: Grade 8

About Science: Grade 8
In level 8, learners study science in its historical, political, and cultural context, and with a growing understanding that science is an ever-changing process rather than a static body of knowledge. Reading level and expectation increase from previous levels.

Science: Grade 8
The Big Picture:
To accomplish the goal of supporting a relationship with the Things of the Universe, a Mason science program consists of nature lore, natural history, and general science. Nature immersion, inquiry, community connection, and supportive literature are woven into each of these three parts. In Form 1, students became acquainted with this area of knowledge, science. Form 2 students extended their scope and began to encounter science as an active part of society. Form 3 students extend their relationship with science in a new direction (time) and consider science as a process by which man seeks Truth within Creation. They begin to situate science in its historical, political, and cultural context with all of the complexities of man's own story. In this way, students are prepared to think about science with a complete and holistic perspective.

Nature lore is timeless knowledge that is passed through a community, much like a grandmother passes on how to make that special bread when the dough just "feels right." Like Mason, we strive to pass on this knowledge primarily through outdoor work. Group nature walks, seasonal readings, and topics in scouting are provided as an Outdoor Work resource in the Quick Links. If desired, literature suggestions to support lore can be found in the Community Read Alouds resource (in Citizenship Grades 4+).

Placement & Combining Tips

About Science: Grade 8
In level 8, learners study science in its historical, political, and cultural context with a growing understanding that science is an ever-changing process rather than a static body of knowledge. Reading level and expectations increase from previous levels.

General Science: Grade 8
For eighth-grade students or possibly hungry seventh graders or ninth graders taking their time. If combining grades is desirable: the combined grades may begin at the appropriate level and move forward together, or teachers can adapt the reading level and laboratory content to meet their needs.

Scheduling

GRADE	SCHEDULE INFO.	BOOKS
8	General Science: Grade 8 3 times/week 30 min	The Story of Science: Einstein Adds a New Dimension The Story of Science: Newton At the Center

Sample Weekly View

Day 1	Day 2	Day 3	Day 4	Day 5
Science: Grade 8				
Natural History: Grade 8	General Science: Grade 8 Nature Walks & Scouting: Grades 1-8	General Science: Grade 8	General Science: Grade 8 Nature Notebook: Grade 8	Labs: Grade 8

Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Science: Grade 8
☐ Obtain any supplies indicated on the science or grade-level supply lists.
☐ Download any apps and shortcut any desired links.

Nature Lore:
☐ Bookmark your Outdoor Work Quick Link, so that you have it available on your weekly outing. Outdoor Work is generally flexible for your location and season and can be moved around in the schedule to incorporate or substitute Natural History Club outings, except where the suggestion is to collect some Thing for a lesson.
☐ Print or bookmark grade-specific nature notebook suggestions to support natural history and general science. Notebooking can be done on a walk, during occupations, or as a field trip, as appropriate.

Special Topics & Field Trips

Science: Grade 8
Encourage students to choose their own special topic and to notice its ecological relationships. What or who are they curious about or interested in getting to know better? Teachers can choose their own special study, too!

☐ Learn a few of your local species or varieties in connection with special topics.

General Science: Grade 8
☐ Term 1: A science museum with motion exhibits
☐ Term 2: A science museum with chemistry exhibits
☐ Term 3: A planetarium or local astronomy club event

Term Prep & Teacher Tips

General Science: Grade 8
☐ Gather household items, typically easy for students to scavenge or teachers to obtain locally:
- 5 Starburst candies or similarly-shaped manufactured item
- sturdy wooden board at least 2-3" wide and 3.5' long to construct marble ramp
- anything to create rails for the ramp, more scrap wood, old pool noodles, etc.
- whatever is needed to attach the rails: bar clamps, hammer and nails, glue, etc.
- books or wood blocks to adjust height
- 3 smaller blocks to act as stops
- 1-6 eggs (Term 1)
- water
- large sheet pan or flat location outside
- 2 plastic cups, 1 smaller and 1 larger
- various objects to test for sliding on ramp, such as small objects made of wood, plastic, metal, etc
- various Lego-type blocks that can be used to build a small box, roughly 1-2" dimensions
- a larger bouncy ball, such as a basketball
- a location to bounce balls
- 1/2 c small solid fill material, such as cat food or rice

- 2 locations at 2 different temperatures, such as a sunny window and a chilly basement
- plastic wrap
- wire cutters, such as those on many pliers
- hammer
- at least 2 different light sources, including bright sunlight and at least 1 type of artificial light
- microwave
- microwave-safe casserole dish, at least 9"
- marshmallows to cover the bottom of the dish, Peeps or jumbo work best
- stopwatch, such as found on most electronic devices
- calculator, such as found on most electronic devices
- printables
- computer
- optional: hot glue gun
- optional: long forceps
- optional: matches
- optional: glass jar that is taller than a tea light candle
- optional: hard boiled egg (Term 2)

Reminders

General Science: Grade 8
☐ Note that Lab in week 6 of Term 1 calls for fresh eggs and Lab in week 2 of Term 2 includes an optional activity that uses a hard-boiled egg. Make a note on your calendar if you will need to purchase these closer to the appropriate dates. Refer to the lab book for more details.

Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.
∞ View Book List Details

General Science: Grade 8


The Story of Science: Einstein Adds a New Dimension


The Story of Science: Newton At the Center

Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.
∞ View Basic Supplies
∞ View Supply List Details

General Science: Grade 8

Household Items - General Science: Grade 8

Quick Links

Click THIS text or scan the QR code for links.

General Science: Grade 8

∞ Extra Helpings

∞ AmScope key for slide IDs

Related Course Quick Links

∞ Outdoor Work

∞ Seek app from iNaturalist

- SkyView Lite for iOS
- SkyView Lite for Android
- Alternate One Year Physical Science Lesson Plans
- Foundations (See Section 13: Science)

General Science: Grade 8

How To Teach

Introduce

• Begin each lesson with learners' existing knowledge. If the book or activity is new or unfamiliar, then look at the title, a picture, or guidance in the lesson plan to help discuss what students think, drawing on previous experience. If continuing or revisiting a topic or activity, then recap.

• Often, the introduction in the lesson plans or the title of the book's section can help learners to draw out the main idea. Use this to support learners, as appropriate.

• Some learners may benefit from using pictures or looking back briefly.

• Allow them time to share any concerns and come alongside, as needed.

Read

• Read or do, as instructed in the lessons, noting any Teacher Tips provided. Learners should always have their journal, notebook, or other paper available in case they need to draw or diagram during the lesson.

• Use supportive strategies and educational tools to reduce frustration and better engage the mind, as appropriate. These could include, but are not limited to, the use of eBooks, pictures, audio, read-aloud, buddy reading, colored reading strips, etc.

• If learners do not understand a word or concept, do not worry. Try to show them with a picture or connect the idea to something they have seen in real life. They are learning much by the way and will likely build understanding over the term, the year, and beyond.

Narrate

• Process the ideas of the lesson by retelling events, describing, explaining a concept, etc. Tips about helping learners deal with various types of passages are provided, but teachers can learn more in Charlotte Mason's School Education Chapter 16, especially p.180.

• Learners may use words, pictures, Legos, PowerPoint, etc, to process and convey ideas in their own way.

• Teachers may take turns to model.

Discuss

• Consider together any thoughts, confusion, or concerns about the passage, keeping in mind that oftentimes new concepts are not going to be 'mastered' on the first or even second introduction. Mastery is not necessarily the goal, but curiosity and thinking.

• Questions/topics for further discussion are often provided in the lesson plans (or even lab books) to help. There are no right or wrong answers to these. Alternatively, many of these can be used for composition, depending on the needs of the learner and the instructional goals of the teacher. If teachers want to keep ideas active in the mind, these can also be used at other times to keep the ideas in the working memory.

• Note that learners may need to spend more than the allotted time engaging with or even repeating a lesson before moving on or as reinforcement at a later time. Adjust the pace as needed to feed the learner.

• Notice if there were any dates that they want to keep for their Book of Centuries.

Connect

● Follow any extra links, examine any sidebars in the text, look at pictures, etc., depending on learner needs and interest.

● These can be viewed as alternative ways to engage.

General Science: Grade 8

Click THIS text or scan the QR code for links.

Term 1

WEEK 1 ⬚ **30m General Science: Grade 8 - Lesson 1**
Setting the Stage

Materials: Newton at the Center

PREP: Read Teacher Tip

→ INTRO
This book tells the story of how physical science developed from the Renaissance into the 1800s. As you move through the story, notice how ideas and knowledge about the world around us changed through a combination of imagination and experimentation. There are many interesting sidebars and tangents in the text, but read the main text first and then explore them, as desired. Watch the course introduction video:
∞ Video Link: Form 3 Science Intro (5:04)

→ READ, NARRATE, & DISCUSS
Ch.1 p.1-6 "For centuries" - "good place to start."

- What do you gather so far about the author's purpose for this chapter?

★ TEACHER TIP
Watch the course introduction video with your student(s).

★ TEACHER TIP
As we will begin some notebooking next week, help students notice the author's structure and purpose this week during discussion.

• IMPORTANT DATES
Beginning of the Renaissance (c.1453)

WEEK 1 ⬚ **30m General Science: Grade 8 - Lesson 2**
Setting the Stage

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.1 p.6-13 "Imagine yourself" - "familiar landmarks."

- This chapter sets the scene for Copernicus. Describe that scene.
- Consider the relationship between church and state in Copernicus' time. Was a singular church and state good for either the church or the person? If so, in what ways? In what ways was it bad for both the church and the person? Is a separate church and state bad for either the church or the person? If so, in what ways? In what ways is it good for both?

★ TEACHER NOTE
A simplified contrast between medieval philosophical and theological beliefs and modern scientific beliefs is presented on the p.10 sidebar. Teachers might allow it to pass by the way, engage students in discussion, or skip this one (the larger ongoing picture takes shape over the year).

WEEK 1 ⬚ **30m General Science: Grade 8 - Lesson 3**
Copernicus

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.2 p.14-20 "Copernicus" - "defines the age."

- Tell about Copernicus' life.
- Describe the shift in ideas observed by the author.

• IMPORTANT DATES
Nicolaus Copernicus (1473-1543)

WEEK 2 ⬚ **30m General Science: Grade 8 - Lesson 4**
Copernicus in the Renaissance

Materials: Newton at the Center

PREP: Read Teacher Tip

→ INTRO
Recall what you read about Copernicus' life and the Renaissance setting

★ TEACHER TIP
Remember to engage students in their reading through natural discussion. If you need help, use the bulleted prompts. Invite them to share their notebook entries to improve the discussion!

• NATURE NOTEBOOK

Term 1

for our story. As you continue reading, begin taking notes, jotting down the ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too. If you'd like to see a few examples, watch the Notebooking video:
∞ Video Link: Notebooking? (1:39)

➔ READ, NARRATE, & DISCUSS
Ch.2 p.20-31 "Since we have" - "and the earth."

- Describe the contradictions observed in Renaissance society.
- Discuss the scientific and religious ideas on Copernicus' mind.

Prompt for General Science in Outdoor Work Quick Link.

WEEK 2

30m General Science: Grade 8 - Lesson 5
Leonardo da Vinci

Materials: Newton at the Center

➔ RECAP
Briefly recall what you read last time.

➔ READ, NARRATE, & DISCUSS
Ch.2 p.32-35 "What's to be" - "and the Arno."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

- Tell about Leonardo da Vinci.

WEEK 2

30m General Science: Grade 8 - Lesson 6
Copernicus Has a Problem

Materials: Newton at the Center

➔ RECAP
Briefly recall what you read last time.

➔ READ, NARRATE, & DISCUSS
Ch.3 p.36-43 "Copernicus is" - "move the earth."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

- What does the author mean by "thought experiment"?
- Discuss some of Copernicus' questions.
- Explain Copernicus's conflict over his ideas.

• IMPORTANT DATES
Copernicus published On the Revolutions of the Heavenly Spheres (1543).

WEEK 3

30m General Science: Grade 8 - Lesson 7
Tycho Brahe Observes

Materials: Newton at the Center

➔ RECAP
Briefly recall what you read last time.

➔ READ, NARRATE, & DISCUSS
Ch.4 p.44-48 "The movement" - "is going on?"
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

- Tell about Tycho's early life.
- Explain why Tycho's approach was such an important development in thought.
- Why did people think that "perfect" meant "unchanging"? Discuss the implications of this idea.

• IMPORTANT DATES
Tycho Brahe (1546-1601)

General Science: Grade 8

[Click THIS text or scan the QR code for links.]

Term 1

WEEK 3

⬚ **30m General Science: Grade 8 - Lesson 8**

Tycho Brahe Observes

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.4 p.48-57 "Only once" - "sending up sprouts."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

- Tell about this episode in Tycho's life.
- Compare the models of Ptolemy, Copernicus, and Tycho using words, drawings, or diagrams.

★ TEACHER NOTE
The timescale of traveling light is mentioned on p.51: "It has taken 20,000 years..." Teachers might choose to allow it to pass by the way or to consider alternative theories they would like to share.

• IMPORTANT DATES
Newton At the Center
Choose events from the timeline on p.56-57.

WEEK 3

⬚ **30m General Science: Grade 8 - Lesson 9**

What is Parallax?

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.4 p.58-61 "When Tycho" - "a solar system."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

- Explain what parallax is and how it is used.